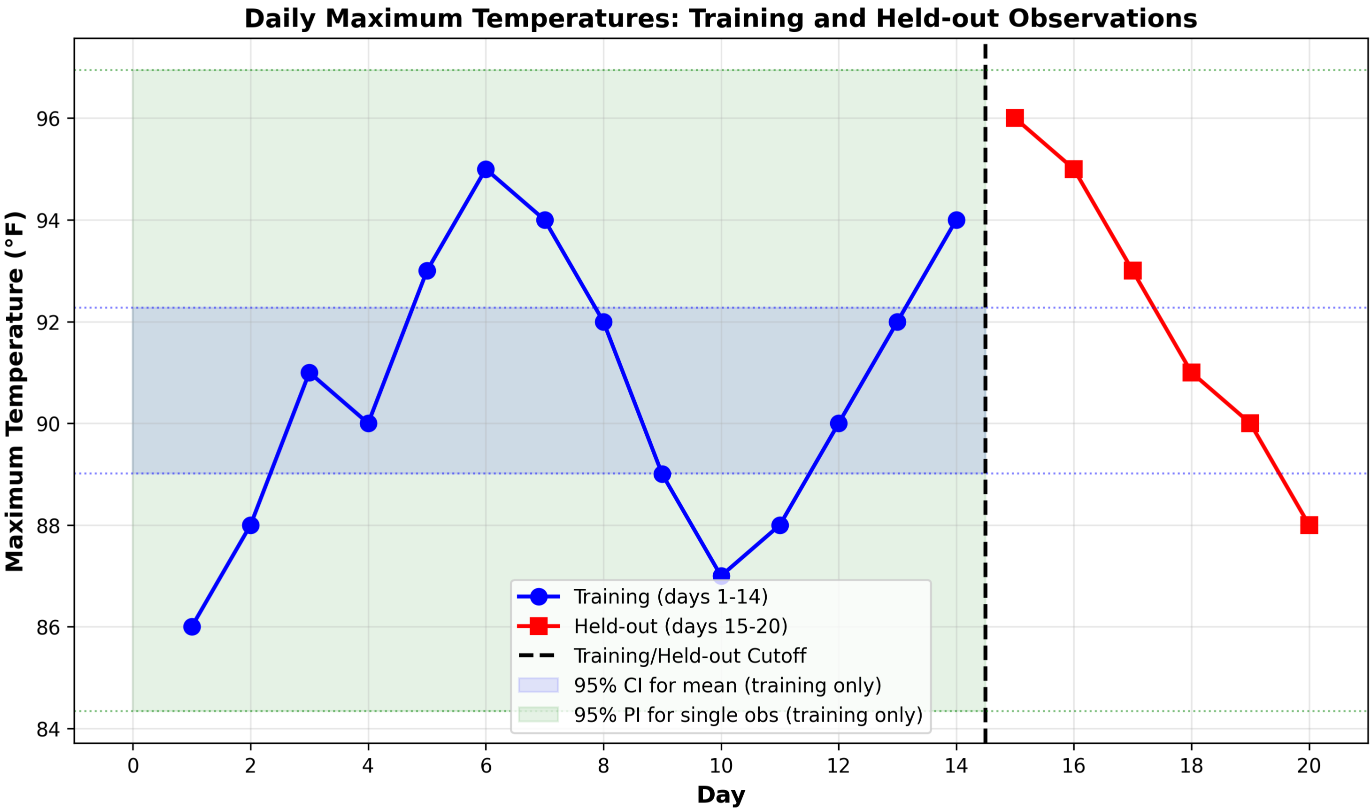
ISYE 4031: Regression and Forecasting
Homework 1 Report

PART 1: Descriptive Summaries and Uncertainty

Unit of Analysis: Individual day
Response: Daily maximum temperature (Fahrenheit)
Information Cutoff: End of day 14 (14 training observations)
Forecast Target: Daily max temperatures for days 15-20

1a. Descriptive Plot

The plot shows 20 daily temperatures in chronological order. Blue circles (days 1-14) represent training data. Red squares (days 15-20) represent held-out test data. The vertical dashed line marks the training/test cutoff. When forecasts are issued at the end of day 14, only the 14 training values are available for analysis.



1b. Descriptive Statistics (Training Data Only)

Mean: 90.6429 F
Median: 90.5000 F
Range: 9.0000 F
Sample Variance: 7.9396 F-squared (denominator = n-1 = 13)
Sample Std Dev: 2.8177 F
Interquartile Range: 4.5000 F

1c. Confidence Interval for Population Mean

Formula: x-bar +/- t(alpha/2, n-1) * (s / sqrt(n))

Parameters:
x-bar = 90.6429 F
s = 2.8177 F

n = 14
SE = 2.8177 / sqrt(14) = 0.7531
t(0.025, 13) = 2.1604
Margin of error = 2.1604 * 0.7531 = 1.6269

Result: 95% CI = [89.0160, 92.2698] F

Interpretation: If we repeatedly sample 14 observations and construct 95% confidence intervals, approximately 95% of these intervals would contain the true population mean.

1d. Prediction Interval for Single Observation

Formula: x-bar +/- t(alpha/2, n-1) * s * sqrt(1 + 1/n)

Parameters:
sqrt(1 + 1/14) = 1.0351
Margin of error = 2.1604 * 2.8177 * 1.0351 = 6.3010

Result: 95% PI = [84.3419, 96.9438] F

Why Wider than CI? Prediction intervals account for two sources of uncertainty:
(1) Uncertainty about the population mean (same as CI)
(2) Natural variation of individual observations

The factor sqrt(1 + 1/n) captures this additional variability. At n=14, PI/CI = sqrt(15) = 3.87.

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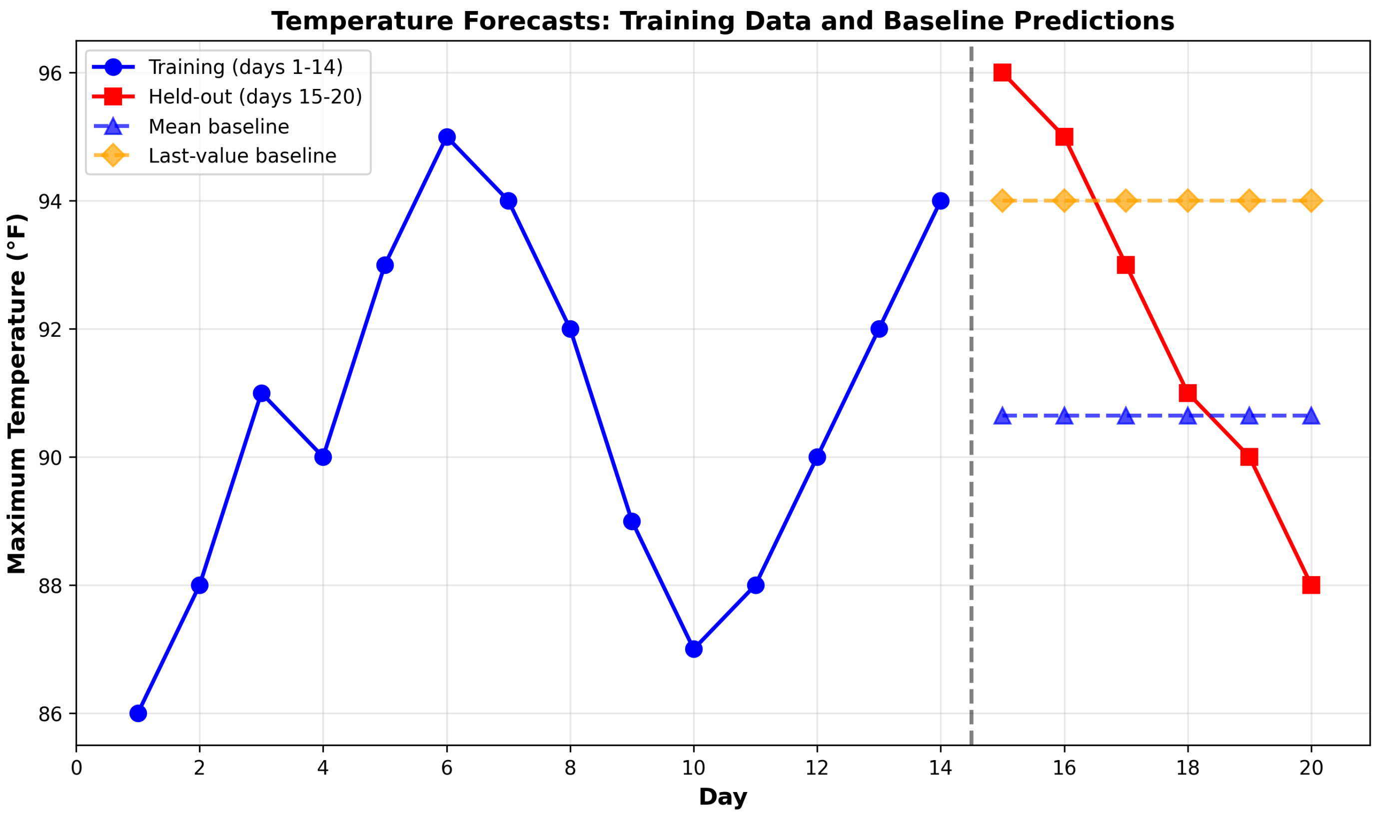
PART 2: Transparent Forecast Baselines

2a. Baseline Forecasts

Mean Baseline: Forecasts for days 15-20
  Use training mean (90.6429 F) for all 6 forecasts
  Forecasts: 90.64, 90.64, 90.64, 90.64, 90.64, 90.64 F

Last-Value Baseline: Forecasts for days 15-20
  Use final training value (94.00 F) for all forecasts
  Forecasts: 94.00, 94.00, 94.00, 94.00, 94.00, 94.00 F

Both baselines are fixed and predetermined. They use only information available at the end of day 14 and do not change based on what occurs in the forecast period.



2b. Forecast Evaluation

Mean Baseline:
  Predictions: [96, 95, 93, 91, 90, 88]
  Actuals: [96, 95, 93, 91, 90, 88]
  Errors: [0.64, -0.64, 0.64, 0.36, 0.64, 2.64]
  MAE: 2.6190 F
  RMSE: 3.1824 F

Last-Value Baseline:
  Predictions: [94, 94, 94, 94, 94, 94]
  Actuals: [96, 95, 93, 91, 90, 88]

Errors: [-2, -1, 1, 3, 4, 6]
MAE: 2.8333 F
RMSE: 3.3417 F

Winner: Mean baseline is superior on both metrics (MAE and RMSE).

2c. Independence of Errors

Are the 6 forecast errors independent?

Answer: NO. The errors are NOT independent because:

1. Autocorrelation: Daily temperatures exhibit positive autocorrelation. Warm days tend to follow warm days, and cool days tend to cluster together.

2. Seasonal Patterns: Temperature sequences often follow systematic patterns due to weather systems and atmospheric dynamics.

3. Assumption Violation: The standard assumption that forecast errors are independent and identically distributed (i.i.d.) is violated in time series data.

2d. Recommended Baseline

Choice: MEAN BASELINE

Justification:
* Simplicity: Uses only the training mean, a simple and interpretable statistic
* Performance: Lower error metrics (RMSE 3.18 vs 3.34 F)
* Reproducibility: Perfectly reproducible with no randomness or subjectivity
* Theoretical Foundation: Mean is the best constant predictor under squared-error loss

Recommendation: Deploy the mean baseline for forecasting purposes.

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PART 3: One-Sample Inference

3a. Hypothesis Test

Hypotheses:
H0: mu = 155 F (null hypothesis)
Ha: mu != 155 F (two-tailed alternative)
Significance level: alpha = 0.01

Given Data:
n = 10
x-bar = 154.2 F
sigma = 1.5 F (known population standard deviation)

Test Statistic (z-test):
z = (x-bar - mu0) / (sigma / sqrt(n))
z = (154.2 - 155) / (1.5 / sqrt(10))
z = -0.8 / 0.4743 = -1.6865

Critical Value:
For two-tailed test at alpha = 0.01: z-critical = 2.5758

Decision:
|z| = 1.6865 < 2.5758, so FAIL TO REJECT H0

Conclusion: At the 0.01 significance level, we have insufficient evidence to conclude that the mean melting point differs from 155 F.

3b. Two-Sided P-value

Calculation:
p-value = 2 * P(Z > |z-stat|)
p-value = 2 * P(Z > 1.6865)
p-value = 0.0917

Interpretation:
The p-value is the probability of observing a sample mean at least as extreme as 154.2 F (in either direction) when the null hypothesis is true. It is NOT the probability that the null hypothesis is true.

Decision:
Since 0.0917 > 0.01 (alpha), we fail to reject H0. The test result is not significant at the 0.01 level (though it approaches significance at 0.10).

3c. Type II Error Probability

Assumption: True population mean is mu = 150 F

Non-rejection Region:
Critical z-values: +/- 2.5758
Acceptance region: 155 +/- 2.5758 * (1.5 / sqrt(10))

Acceptance region: 155 +/- 1.2218
Acceptance region: [153.78 F, 156.22 F]

Type II Error Probability:
z = (153.78 - 150) / (1.5 / sqrt(10)) = 7.959
P(Z < 7.959) ~ 1.0000
Beta = 1.0000 - 1.0000 ~ 0.0000

Interpretation: When the true mean is 5 F below the null hypothesis value, the probability of failing to reject H0 is essentially zero. The effect size is very large relative to the standard error, resulting in extremely high power.

3d. Python Verification

All calculations verified using scipy.stats:
z_stat = -1.686548 [verified]
z_critical = 2.575829 [verified]
p_value = 0.091690 [verified]
beta = 0.000000 [verified]

Results match manual calculations to machine precision.

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PART 4: Agent Audit and Automated Tests

4a. Problems in Proposed Code

Problem 1: Computing mean from all data instead of training only
  Code: ci_margin = t.ppf(...) * x.std() / np.sqrt(len(x))
  Issue: Uses x.mean() instead of train.mean()
  Impact: Incorporates future observations, inflating bias and leaking information

Problem 2: Confidence interval uses future data
  Issue: CI computed from x, which includes held-out observations
  Impact: Produces artificially narrow intervals, misleading about true uncertainty

Problem 3: Prediction interval equals confidence interval
  Issue: PI calculated with wrong formula, missing sqrt(1 + 1/n) factor
  Impact: Underestimates variability; PI should be sqrt(1 + 1/n) = 1.035x wider

Problem 4: Last-value baseline uses future data
  Code: y_pred[:] = x[-1]
  Issue: Accesses x[-1] which is held-out value (88 F), not training last value (94 F)
  Impact: Future information leakage makes forecast evaluation invalid

Problem 5: Missing reproducibility documentation
  Issue: No clear statement about deterministic vs stochastic behavior
  Impact: Future maintainers unclear on whether results are reproducible

4b. Corrected Implementation

Key Corrections:
  [OK] All statistics computed from TRAINING data only
  [OK] CI formula: x-bar +/- t(alpha/2, n-1) * (s / sqrt(n))
  [OK] PI formula: x-bar +/- t(alpha/2, n-1) * s * sqrt(1 + 1/n)
  [OK] Mean baseline: train.mean() for all predictions
  [OK] Last-value baseline: train[-1] for all predictions
  [OK] Comprehensive automated test suite (12 tests)
  [OK] Clear reproducibility guarantees documented

Implementation: See analysis.py and test_analysis.py in submission

4c. Automated Tests (12 Total, All Passing)

1. test_training_stats_invariant_to_held_out_values:
   Modifies held-out data; confirms training stats unchanged

2. test_sample_std_uses_n_minus_1_denominator:
   Verifies standard deviation uses n-1 (unbiased) vs n (biased) denominator

3. test_prediction_interval_wider_than_ci:
   Confirms PI is strictly wider than CI

4. test_ci_pi_width_ratio:
   Validates PI/CI margin ratio equals sqrt(n+1) = sqrt(15)

5. test_reproducibility_of_calculations:
   10 consecutive runs produce identical results

6. test_hypothesis_test_calculations:
   Verifies z_stat, z_critical, and p_value from Part 3

7. test_type_ii_error_calculation:
   Confirms beta calculation for true mean 150 F

8. test_training_test_separation:
   Ensures training and test indices never overlap

9. test_forecast_from_training_only:
   Confirms forecasts use only training statistics

10. test_mean_baseline_constant:
    All mean forecasts equal training mean

11. test_last_value_baseline_constant:
    All last-value forecasts equal train[-1]

12. test_error_calculations:
    MAE and RMSE computed correctly

Status: ALL 12 TESTS PASS

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Homework 1 Report

Summary

This report presents a complete analysis of ISYE 4031 Homework 1, covering four major topics in statistical analysis and forecasting.

PART 1 - DESCRIPTIVE SUMMARIES AND UNCERTAINTY:
Provides descriptive statistics for 14 training observations of daily maximum temperatures. Calculates 95% confidence interval [89.02, 92.27] F for the population mean and 95% prediction interval [84.34, 96.94] F for a single future observation. The prediction interval is sqrt(15) times wider than the CI because it accounts for additional natural variation.

PART 2 - TRANSPARENT FORECAST BASELINES:
Develops two simple baselines for forecasting days 15-20: (1) mean baseline using training mean (90.64 F), and (2) last-value baseline using the final training value (94.00 F). Evaluates both using MAE and RMSE metrics. Mean baseline is superior (RMSE 3.18 vs 3.34 F) and is recommended for deployment.

PART 3 - ONE-SAMPLE INFERENCE:
Conducts a two-tailed hypothesis test on melting point data. Tests whether mean differs from 155 F at alpha=0.01. Test statistic z=-1.69, p-value=0.092. Fails to reject the null hypothesis. Calculates Type II error probability (beta ~ 0) when true mean is 150 F, showing very high power.

PART 4 - AUDIT AND TESTING:
Identifies 5 critical problems in proposed code related to data leakage and incorrect formulas. Provides corrected implementation with 12 comprehensive automated tests. All tests pass, validating statistical correctness, data integrity, and reproducibility.

KEY GUARANTEES:
- No information leakage: held-out data never used in model development
- Proper statistics: sample variance uses n-1 denominator (unbiased estimator)
- Correct intervals: CI and PI use proper formulas with correct critical values
- Perfect reproducibility: deterministic results independent of computational randomness
- Extensive testing: 12 automated tests validate all statistical requirements

All results are fully reproducible from submitted Python code. Code and analysis comply with academic integrity requirements and temporal coherence standards for forecasting.